Sudhanshu Sakhala

+91-7744869769 sudhanshu.sakhala2002@gmail.com linkedin.com/in/sudhanshu-sakhala-800444191

PROFESSIONAL SUMMARY

AI Engineer with 1.8+ years of experience architecting and deploying production-grade Agentic AI systems, LLM applications, and enterprise AI platforms. Strong expertise in LangGraph, LangChain, MCP, RAG, LLM orchestration, context engineering, AI memory, and PostgreSQL/pgvector. Experienced in integrating 6+ LLM providers, building 150+ MCP tools, and developing scalable FastAPI-based agent services with streaming, observability, and reliability controls. Proven track record of reducing context usage by ∼93%, token consumption by 65%, and LLM inference costs by 70–80% through intelligent architecture and optimization.

EDUCATION

Dr. Vishwanath Karad MIT World Peace University Aug 2022 – Aug 2024
Master of Computer Application (MCA) Pune, India

- **CGPA:** 8.11 / 10

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, PySpark, Bash, JavaScript, TypeScript

AI & LLM Engineering: Agentic AI, LangChain, LangGraph, MCP, ReAct Agents, RAG, LLM Orchestration, Tool Calling, Prompt Engineering, Context Management, Prompt Caching, Token Optimization, OpenAI GPT, Anthropic Claude, Gemini, Qwen

AI Infrastructure & Evaluation: Langfuse, LLM Observability, Golden Dataset Evaluation, LLM Routing, Failover, SSE Streaming, Context Isolation, Cost Optimization

Backend Development: FastAPI, REST APIs, AsyncIO, Pydantic, Microservices, Authentication, Authorization, RBAC, Rate Limiting, Session Management, SSE

Data & Retrieval: PostgreSQL, pgvector, HNSW, BM25, RRF, Cross-Encoder Reranking, Knowledge Graphs, Redis, MongoDB

Cloud Platforms: Microsoft Azure (ADF, ADLS Gen2, Azure Databricks, Azure Synapse Analytics), Google Cloud Platform (GCP), AWS, Yeedu

DevOps & Tools: Docker, Kubernetes, Azure DevOps, CI/CD, Git, Linux, Portkey, JSON

PROFESSIONAL EXPERIENCE

Modak Analytics (ForgeAI) Nov 2024 – Present
AI Engineer Hybrid

- Architected and shipped ForgeAI, a production AI platform managing 5 MCP servers and 130+ tools integrating 5+ enterprise data sources (GitHub, Jira, Yeedu, Databricks, Reltio) through a unified AI interface.
- Designed a multi-stage LangGraph agent system (Analyst → Data Engineer → Tester) with dynamic tool filtering and context-isolated sub-agents, reducing parent context from 757K to 50K tokens (∼93% reduction).
- Built high-performance RAG and memory systems using PostgreSQL, pgvector, HNSW, BM25, RRF, Cross-Encoder reranking, and knowledge-graph retrieval, achieving sub-300ms query latency.
- Engineered a 5-layer context optimization pipeline with summarization, context editing, tool-result truncation, progressive discovery, and repetition guards, reducing token usage by 65% and enabling 4× larger effective context windows.
- Integrated 6 LLM providers (OpenAI, Claude, Gemini, Qwen, Azure OpenAI, Grok) via Portkey with intelligent routing, failover, and prompt caching, achieving 70–80% LLM cost reduction.

– Built production-grade AI infrastructure with Langfuse observability, Golden Dataset evaluations, SSE streaming, Redis concurrency controls, conversation branching, RBAC, prompt-injection protection, and zero-storage credential forwarding.

Modak Analytics (Yeedu.io)
AI Engineer Hybrid

– Designed and developed Yeedu AssistantX, a production-grade AI assistant for cloud platform operations and developer workflows.
– Built a provider-agnostic LLM orchestration framework supporting OpenAI GPT and Anthropic Claude using LangChain and LangGraph.
– Developed scalable FastAPI microservices with streaming responses, authentication, rate limiting, and Redis-based session management.
– Created and integrated 25+ MCP tools enabling AI agents to interact with clusters, jobs, notebooks, Git repositories, storage systems, and metadata services.
– Implemented prompt caching, token tracking, and conversation summarization, reducing inference costs and improving context efficiency.
– Containerized applications using Docker and enhanced agent reliability through structured error handling, tool recovery, and execution safeguards.

ACHIEVEMENTS

Built and deployed ForgeAI, a production-grade enterprise AI platform integrating 5 MCP servers, 130+ tools, and 5+ enterprise data sources for agent-driven automation.

Engineered multi-agent LangGraph, RAG, memory, and context optimization systems, reducing agent context consumption by $\sim$93%, token usage by 65%, and achieving sub-300ms retrieval latency.

Integrated 6 major LLM providers with intelligent routing, failover, and prompt caching, achieving 70–80% reduction in LLM inference costs.

Developed Yeedu AssistantX with 25+ MCP tools, multi-provider LLM orchestration, FastAPI microservices, streaming, authentication, and Redis-based session management.